

Layers *of* Meaning

ABOUT THE PROJECT

This project is all about comparison. We study broadsides (everyday, street-level prints) and publications (more formal, often official texts) from the National Library of Scotland. Here on this website, you can explore two kinds of comparisons:

HOW IT WORKS

We clean and structure the data, define historical topic categories, then use a large language model via API to classify. We classify broadsides twice (by title and by full content) to compare the

FURTHER EXPLORATION

LEGENDS

- broadsides
- publication

Layers Of Meaning

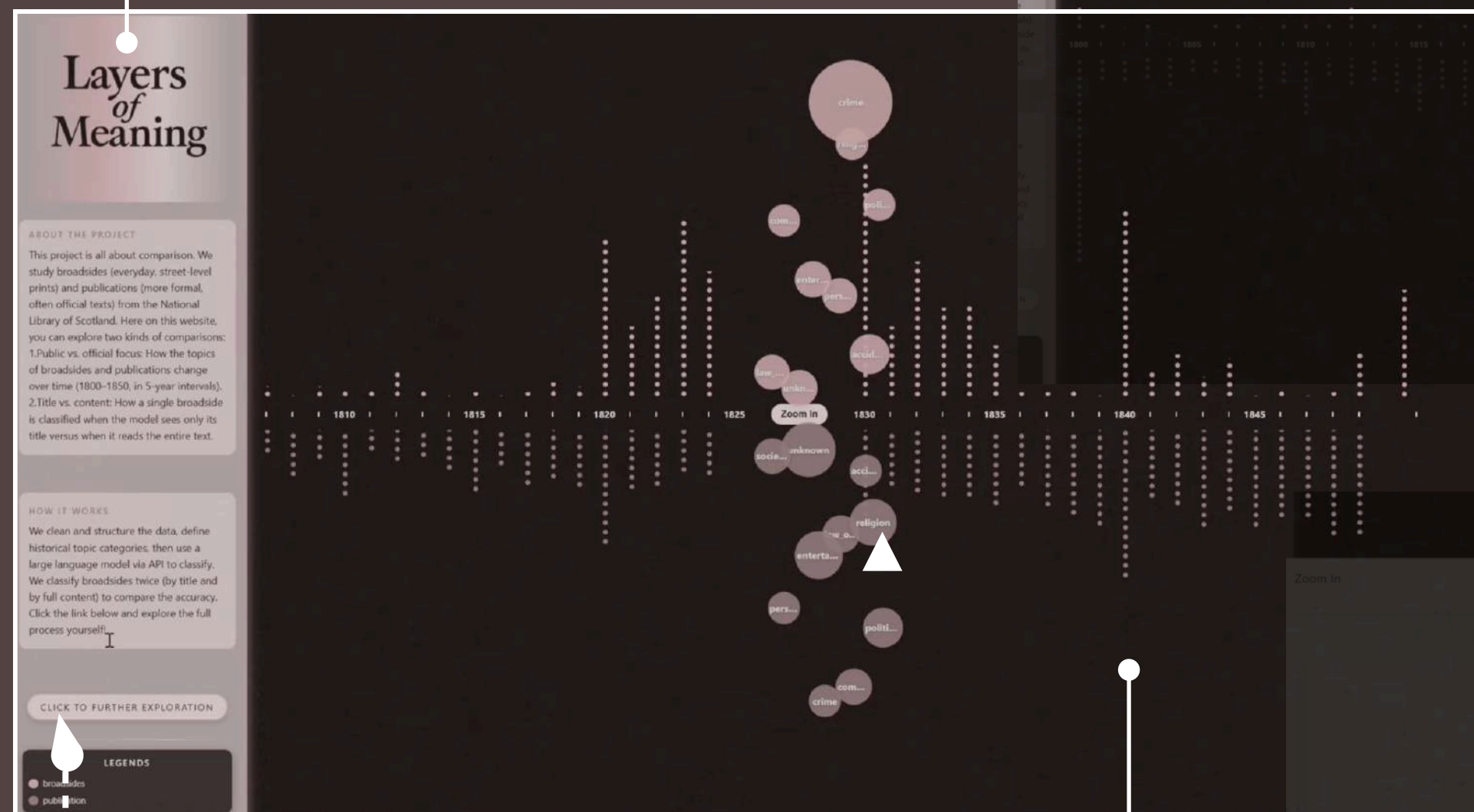
An interactive website to explore the differences between official publications and public broadsides in 1800-1850 based on *National Museum of Scotland database*.

Demo video: <https://youtu.be/mw0BRKU2h9o?si=3zQ5htFdNoSL9hHi>

INTERFACE PREVIEW

Navigation Page

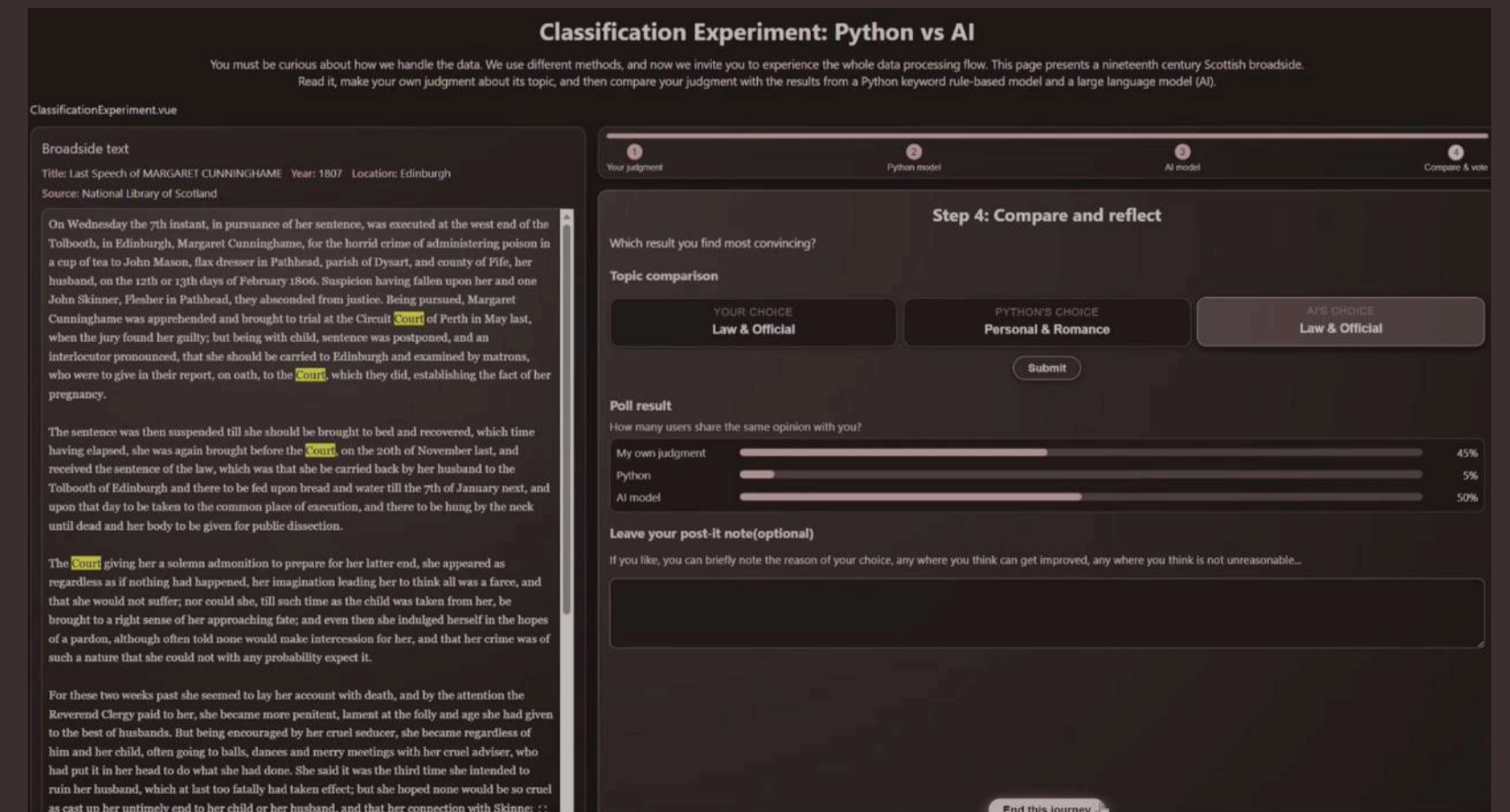
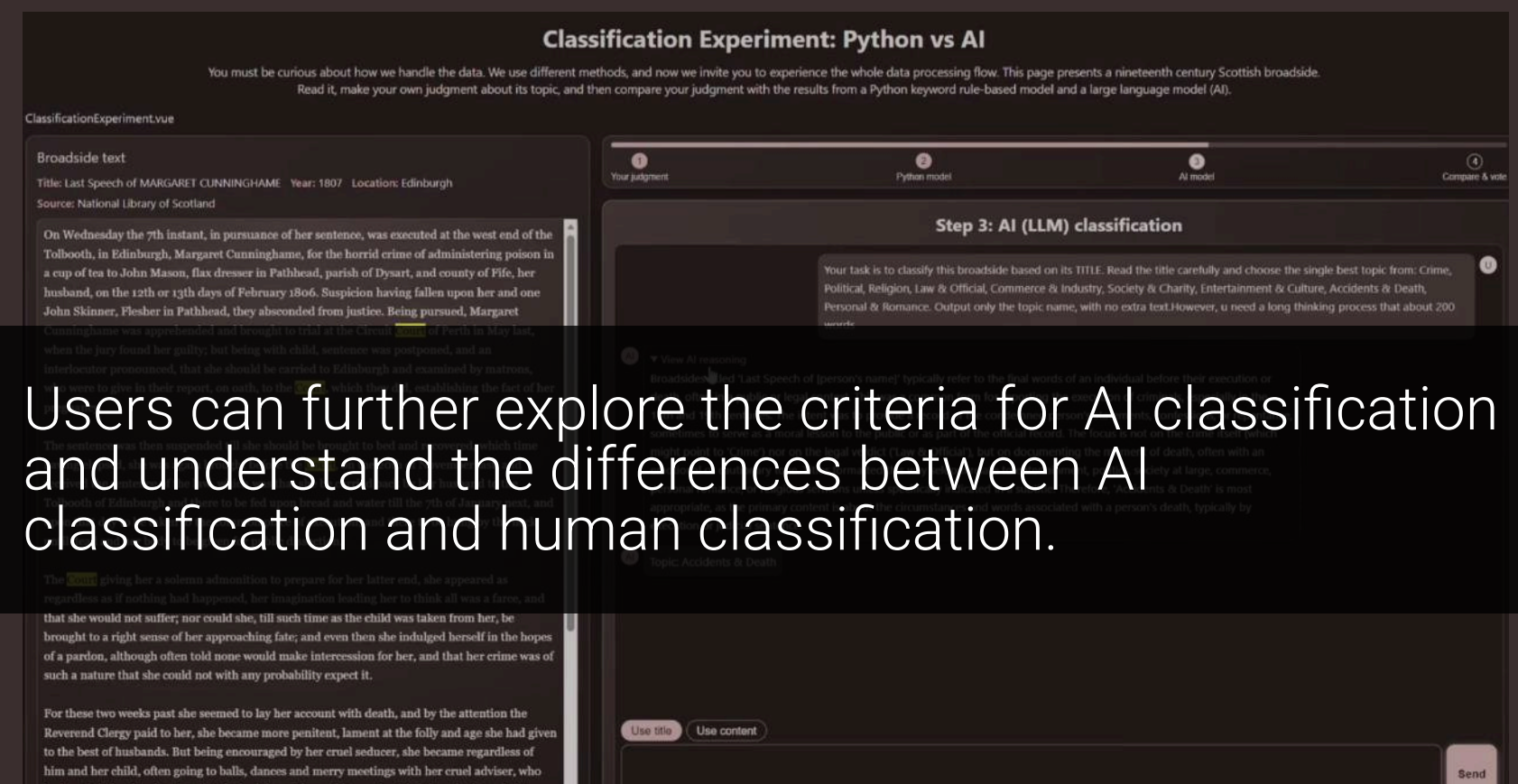
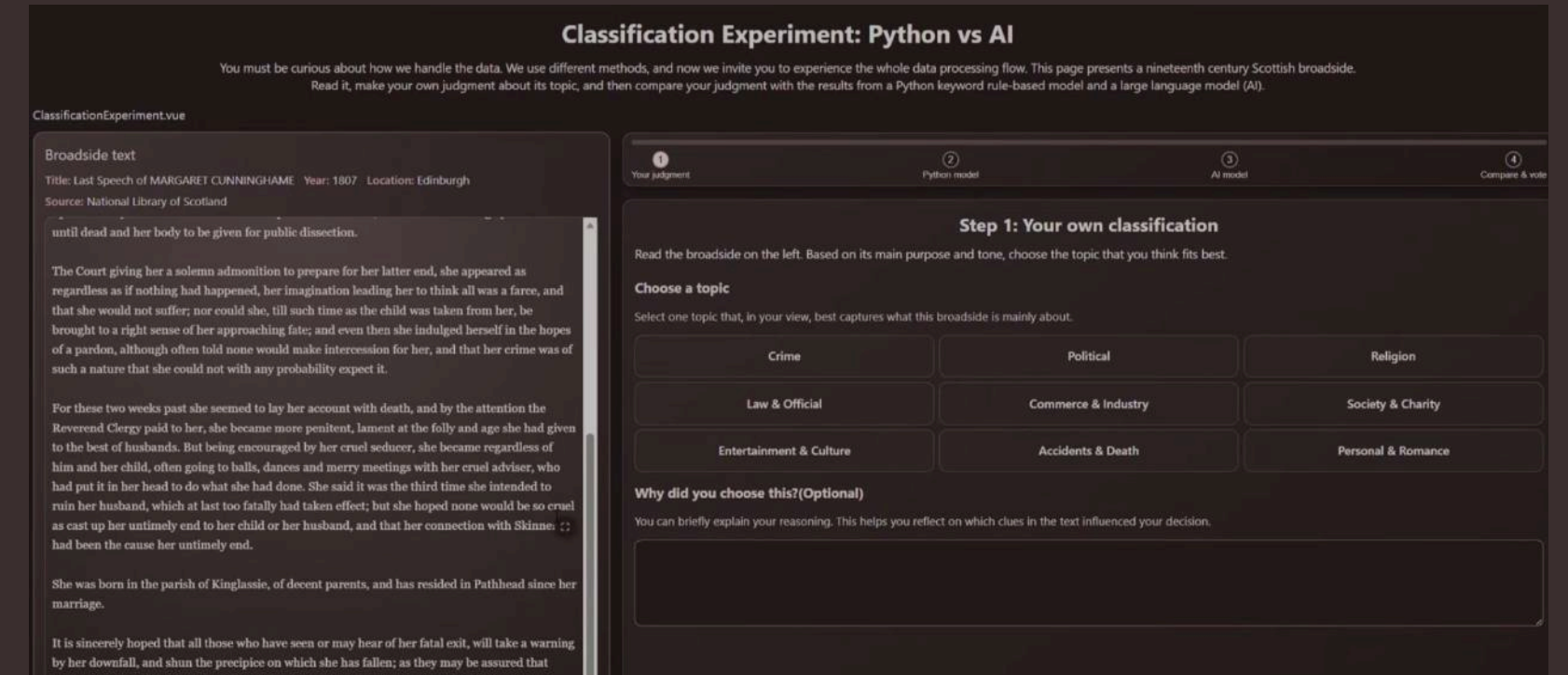
Introduction to the two types of datasets and project background



Homepage

The timeline displays the thematic classification results of official publications and street broadsides.

Click on the topic category results to view detailed information for each publication or broadsides.



Demo video: <https://youtu.be/mw0BRKU2h9o?si=3zQ5htFdNoSL9hHi>

APPENDIX: DATA PROCESSING

Design Comparative Testing for LLM and Python

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # read excel file
5 df = pd.read_excel("/Users/chenzizhan/Desktop/ai_title.xlsx")
6
7 # count the quantity of each topic of ai_title
8 theme_counts = df["ai_title"].value_counts()
9
10 # draw a bar chart
11 plt.figure(figsize=(10, 6))
12 plt.bar(theme_counts.index, theme_counts.values)
13 plt.xlabel('Themes', fontsize=12)
14 plt.ylabel('Count', fontsize=12)
15 plt.title('LLM Classification Results Based on the title of BroadSides', fontsize=14)
16
17 plt.xticks(rotation=45, ha='right')
18
19 plt.tight_layout()
20 plt.show()
21

```

```

/Users/chenzihan/python_work/.venv/bin/python /Users/chenzihan/python_work/library coursework/ai_title pr
2025-11-28 00:50:50.228 Python[36874:3297211] *([IMKClient subclass]: chose IMKClient_Modern
2025-11-28 00:50:50.228 Python[36874:3297211] *([IMKInputSession subclass]: chose IMKInputSession_Modern

```

Figure 2: Bar chart code

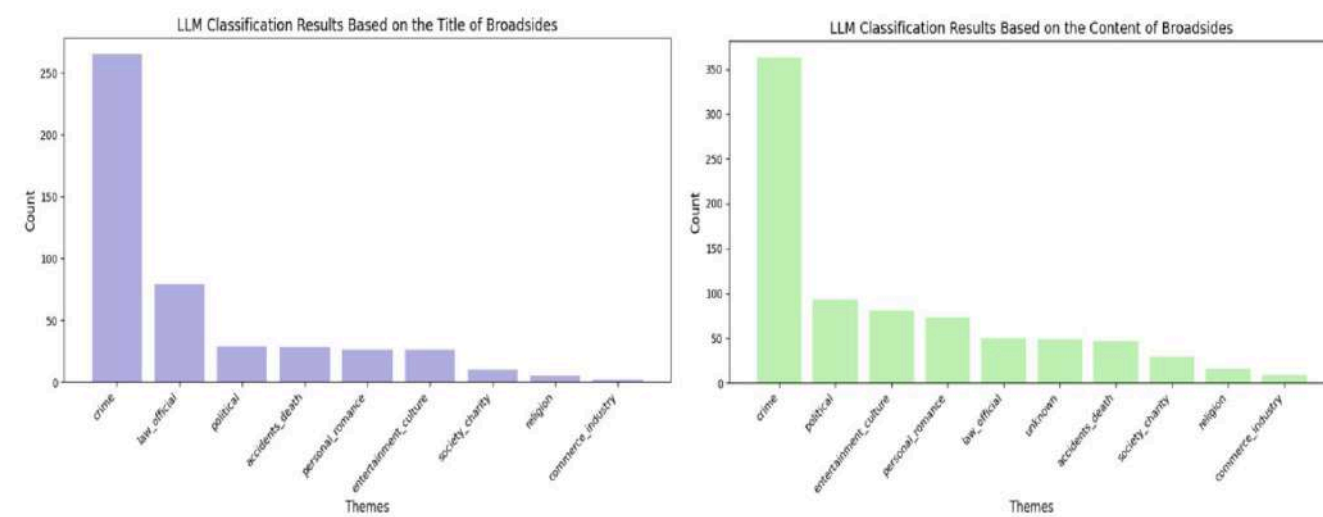


Figure 3: Classification results count

Human Sampling Verification

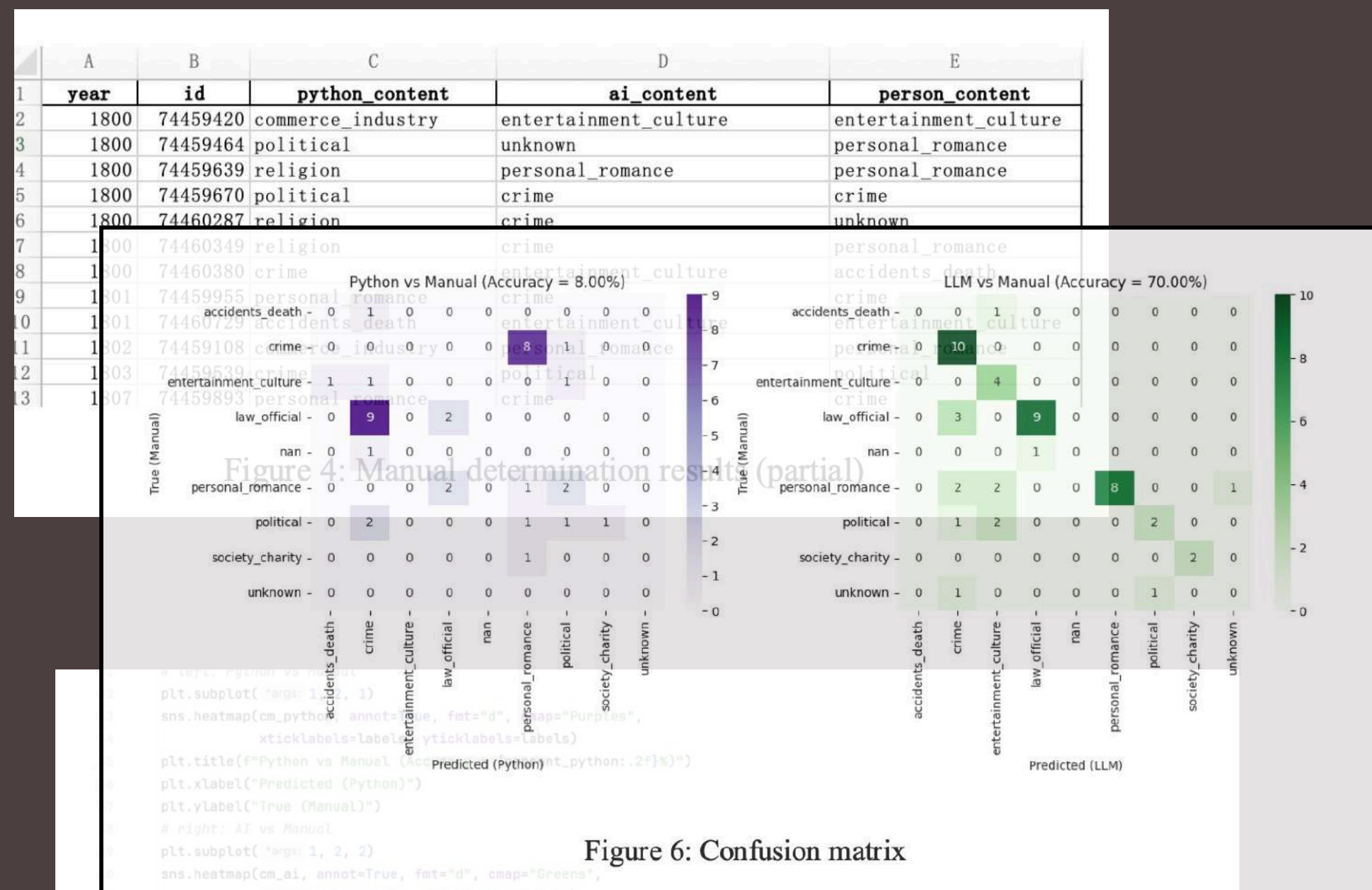


Figure 6: Confusion matrix

```

42 plt.title(f"AI vs Manual (Accuracy = {percent_ai:.2f}%)")
43 plt.xlabel("Predicted (AI)")
44 plt.ylabel("True (Manual)")
45
46 plt.tight_layout()
47 plt.show()
48

```

Manual comparison

```

/Users/chenzihan/python_work/.venv/bin/python /Users/chenzihan/python_work/library coursework/Manual comparison.py
'person_content' is the same percentage as 'person_content': 8.00%
'ai_content' is the same percentage as 'person_content': 70.00%
2025-11-23 23:30:10.872 Python[9571:547865] +[IMKClient subclass]: chose IMKClient_Modern
2025-11-23 23:30:10.873 Python[9571:547865] +[IMKInputSession subclass]: chose IMKInputSession_Modern

```

Figure 5: Matrix code

Secondary Verification

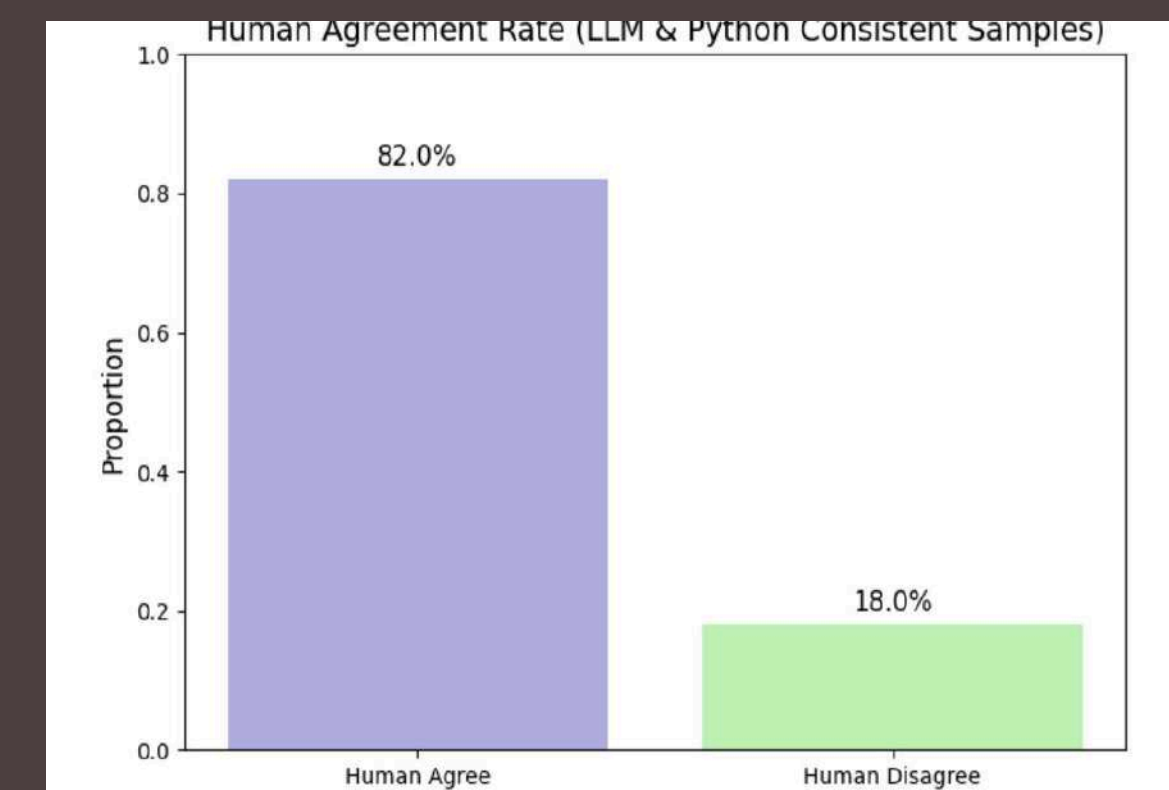


Figure 7: Human agreement rate

Data Logic Reasoning

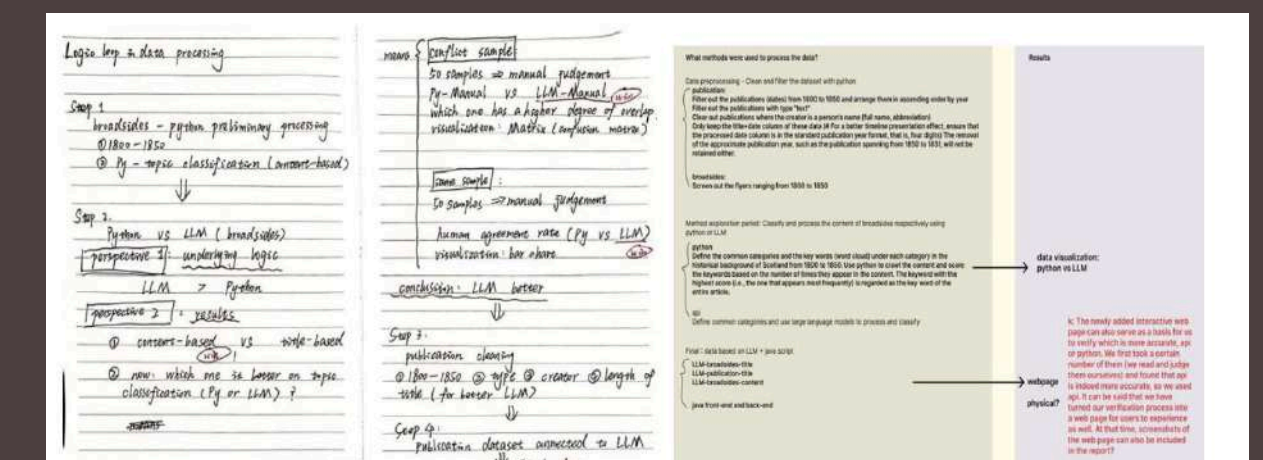


Figure 8: Logical framework sketch

Figure 9: Process sorting